Machine Learning Foundations

Supervised Learning, Data Splits, and Scikit-Learn

Features X + labels y → fit a model → evaluate honestly → understand failure modes

WEEK 11 • LECTURE 22 • 75 MINUTES

Learning outcomes

BY THE END OF LECTURE 22

00–20 MIN

- Write the supervised-learning problem as a mapping from feature vectors to labels.
- Explain the distinct roles of training, validation, and test data.
- Recognize source-level leakage in segmented audio datasets.
- Convert time-varying MFCC matrices into fixed-length vectors for classical models.
- Train and evaluate a baseline classifier using the scikit-learn estimator API and a confusion matrix.

From rules to learned mappings

SUPERVISED LEARNING CHANGES WHO WRITES THE DECISION RULE

00–20 MIN

Traditional program
Engineer writes rules
Input + rules → output
Useful when rules are explicit

Supervised ML
Provide labeled examples
Algorithm estimates θ
Generalize to unseen examples

Audio example
x = audio features
y = class label
$\hat{y}$ = predicted class

$$\hat{y} = f(x\,;\,\theta)$$

Feature matrix X and target vector y

SCIKIT-LEARN USES A TABULAR CONTRACT

00–20 MIN

$$X \in R^\wedge(N \times D) \qquad y \in \{1,\ldots,C\}^\wedge N$$

N
number of clips

D
features per clip

C
number of classes

- Every row X[i,:] must have the same length D.
- y[i] is the label paired with exactly that row.
- For 20-MFCC mean+std features, D = 40.

What is a “feature”?

A COORDINATE CHOSEN TO MAKE PREDICTION EASIER

00–20 MIN

- Raw waveform sample: physically direct, but high dimensional and alignment sensitive.
- Mel bin at one time frame: localized perceptual-frequency energy.
- MFCC coefficient: compact combination of log-Mel bands.
- Time-mean MFCC: one number summarizing a whole clip.
- Feature engineering chooses which invariances the model gets for free.

Invariance example
Time averaging makes the representation less sensitive to WHEN a sound occurs—but it also hides temporal order.

Train, validation, and test

THREE DIFFERENT JOBS

20–40 MIN

TRAIN
Fit model parameters
Fit data-dependent preprocessing
Seen throughout development

VALIDATION
Choose hyperparameters
Compare feature/model choices
May guide development

TEST
Final unbiased estimate
Do not tune against it
Open after choices are fixed

Test set $\neq$ another validation set

The golden rule for audio: split by source, not just by clip

SEGMENTS FROM ONE PHYSICAL RECORDING ARE CORRELATED

20–40 MIN

Source recording A

clip 1 | clip 2 | clip 3 | clip 4

Source recording B

clip 1 | clip 2 | clip 3 | clip 4

BAD: random clips can place A1 in train and A2 in test

GOOD: all clips from a source stay in one split/fold

Instructor cue: Use the siren-segmentation story as a hypothetical. The central problem is source identity leaking across splits.

UrbanSound8K: honor the predefined folds

FOLD IDENTITY IS METADATA, NOT A NUISANCE COLUMN

20–40 MIN

- Use the dataset-provided fold assignments instead of blindly calling train_test_split on excerpts.
- For one course baseline, a simple convention is folds 1–8 train, 9 validation, 10 test.
- For stronger estimates, rotate held-out folds and report cross-validation results.
- Keep the split rule fixed while comparing feature representations.

Course split
1–8 TRAIN

9 VALIDATION

10 TEST

Leakage can occur in preprocessing too

FIT STATISTICS ON TRAINING DATA ONLY

20–40 MIN

$$z_j = (x_j - \mu_{j,train}) / \sigma_{j,train}$$

- If μ and σ are computed using test clips, information from the test distribution leaks into training.
- scikit-learn Pipeline helps fit transformers only on the training fold during model fitting.
- Random forests do not require standard scaling, but SVM/logistic models often benefit from it.

```
pipe = Pipeline([
 ('scale', StandardScaler()),
 ('clf', SVC(kernel='rbf')),
])
pipe.fit(X_train, y_train)
```

2D audio features meet 1D classical models

WE MUST CHOOSE HOW TO REMOVE THE TIME AXIS

40–55 MIN

MFCC matrix [20 × 173]	**Flatten** [3460]	**Mean only** [20]	**Mean + std** [40]	**CNN later** keep 2D

- Flatten: keeps frame order but creates many coordinates and assumes fixed alignment.
- Mean: very compact but ignores temporal ordering and dynamics.
- Mean + standard deviation: simple compromise for a classical baseline.
- CNNs can consume the 2D map and learn local time–frequency patterns.

Time averaging is a model assumption

WHAT INVARIANCE DID WE JUST HARD-CODE?

40–55 MIN

$$\bar{c}[b] = (1/T)\ \Sigma\_t\ C[b,t]$$

- The average keeps "how much" of each coefficient occurred.
- It discards "when," "in what order," and "how long each pattern persisted."
- A rising siren and a falling siren can have similar averages.
- This is exactly the kind of ceiling that motivates sequence models and CNNs.

Compression buys simplicity by spending structure.

The scikit-learn estimator API

SAME MENTAL MODEL ACROSS MANY ALGORITHMS

55–75 MIN

```
clf = RandomForestClassifier(
    n_estimators=300,
    random_state=42,
    n_jobs=-1,
)

clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
accuracy = clf.score(X_test, y_test)
```

- Instantiate: choose model family and hyperparameters.
- fit: estimate model from training data.
- predict: produce labels for unseen feature vectors.
- score/metrics: quantify performance—then inspect errors.

Why Random Forest is a useful first baseline

NOT NECESSARILY BEST—EXCELLENT FOR DIAGNOSIS

55–75 MIN

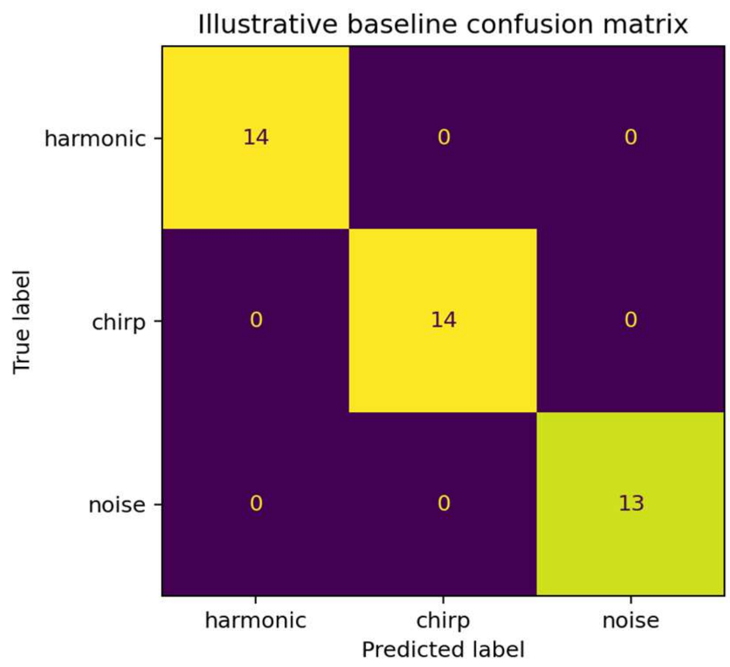
- Handles nonlinear decision boundaries without feature scaling.
- Works naturally with compact tabular summary features.
- Fast to train compared with a deep network.
- Provides a sanity check: are the engineered features carrying class information at all?
- Its accuracy is dataset/split/feature dependent—do not promise a fixed 60–65% ceiling.

Baseline ≠ target
A baseline should be reproducible, interpretable, and difficult to accidentally leak—not artificially impressive.

Confusion matrix: accuracy with structure

WHICH CLASSES ARE BEING CONFUSED WITH WHICH?

55–75 MIN



- Rows: true classes; columns: predicted classes in the displayed convention.
- Diagonal cells: correct classifications.
- Off-diagonal cells reveal systematic confusions.
- The example shown is synthetic only—it validates the workflow, not UrbanSound8K performance.

Accuracy is not the whole evaluation

USE CLASS-SENSITIVE METRICS AND ERROR ANALYSIS

55–75 MIN

$$\text{accuracy} = (\#\ \text{correct}) \,/\, N$$

Precision | **Recall** | **Macro-F1**

- Class imbalance can make overall accuracy misleading.
- Macro-averaging gives each class equal weight.
- Inspect representative false positives and false negatives in the original audio.
- Ask whether errors come from the model, the feature representation, or preprocessing.

Live demo: build X and y without leakage

METADATA DRIVES THE SPLIT

55–75 MIN

```
meta = pd.read_csv(root / 'metadata' / 'UrbanSound8K.csv')

train = meta[meta['fold'].isin(range(1, 9))]
valid = meta[meta['fold'].eq(9)]
test  = meta[meta['fold'].eq(10)]

X_train, y_train = featurize_rows(train, root)
X_valid, y_valid = featurize_rows(valid, root)
X_test,  y_test  = featurize_rows(test, root)
```

Instructor cue: Ask why splitting metadata first is safer than extracting all clips and shuffling later.

A robust baseline feature: MFCC mean + standard deviation

SMALL D, FAST EXPERIMENTS

55–75 MIN

```
mfcc = librosa.feature.mfcc(S=mel_db, n_mfcc=20)
feat = np.concatenate([
    mfcc.mean(axis=1),
    mfcc.std(axis=1),
])
# feat.shape == (40,)
```

- Mean captures typical spectral-envelope coordinates.
- Std captures how much those coordinates vary over time.
- Still destroys sequence order, but usually stronger than mean alone.
- A clean baseline lets next week’s CNN earn its complexity.

Reproducibility is part of engineering

A BASELINE SHOULD BE RERUNNABLE

55–75 MIN

- Freeze the split policy and record feature parameters.
- Set random_state for stochastic estimators.
- Cache extracted features when iteration becomes slow.
- Report class labels, fold assignments, and metrics alongside accuracy.
- Never tune on the test fold because “it is only a classroom project.”

Reproducibility checklist
sample rate
duration
n_fft / hop
n_mels / n_mfcc
split folds
model seed

Error analysis: listen to your mistakes

THE CONFUSION MATRIX TELLS YOU WHERE TO LOOK

55–75 MIN

- Find high-confidence wrong predictions.
- Listen to the clip and inspect its Mel-spectrogram.
- Check label ambiguity, background sounds, clipping, silence, and preprocessing artifacts.
- Compare two representation choices on the same failed example.
- A better feature may fix the problem more cheaply than a bigger model.

ML engineering loop
measure → inspect → hypothesize → change one thing → remeasure

Why can a classical baseline hit a ceiling?

REPRESENTATION MAY BE THE BOTTLENECK

BRIDGE

Mean MFCC → Random Forest → Decision

- Time averaging removes trajectories, modulation patterns, and event shape.
- Flattening keeps order but makes the model sensitive to exact alignment and creates thousands of coordinates.
- A CNN can treat the Mel-spectrogram as a 2D field and learn local patterns that repeat at different times.

Next week: preserve the picture, learn the filters.

Concept check

WHICH EVALUATION IS TRUSTWORTHY?

CHECK

- A. Randomly split all segmented clips, then standardize using the full dataset.
- B. Use source-aware predefined folds; fit preprocessing and model on training only; test once at the end.
- C. Tune hyperparameters until test accuracy stops improving.
- D. Report only training accuracy because the labels are known to be correct.

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Correct: B

Lecture 22 synthesis

A TRUSTWORTHY BASELINE IS A SCIENTIFIC INSTRUMENT

WRAP-UP

- Supervised learning learns f(x;θ) from labeled feature vectors.
- Data splitting is part of the model: leakage can make an invalid system look excellent.
- Classical models need fixed-length vectors, forcing an explicit choice about time structure.
- scikit-learn makes model APIs consistent; honest evaluation still depends on us.
- The baseline establishes what compact hand-engineered features can do before CNNs see the full 2D representation.

Exit ticket: Why can a more accurate random clip split be a worse scientific result?